J. Conor Schehl

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EDUCATION

Colorado School of Mines, Golden, CO | BS, Electrical Engineering | GPA: 4.0 | Intent on MS in IC & Mixed-Signal Design | **Expected Graduation (Undergrad): May 2028**

- **Relevant Coursework:** Circuit Analysis, Digital Logic (in progress), Feedback (in progress), Embedded Systems (upcoming), Electronic Devices (upcoming)

SKILLS

- **Hardware**: PCB Design (Altium), Schematic Capture, PCB Bring-up, STM32, SMD/SMT Soldering, Circuit Analysis, LTspice, JTAG, Oscilloscopes, Hardware Debugging
- **Embedded**: Python, C/C++, CAN, SPI, I^2C, UART, Linux, MATLAB, Digital Logic
- **General**: Microsoft Suite, Team Collaboration, Design Reviews, SolidWorks, DFR/DFM

EXPERIENCE

Formula Society of Automotive Engineers | Electrical Engineer | **September 2025 - Present**

- Collaborate with a multidisciplinary team to make formula-style racecars perform competitively at the collegiate level
- Design and assemble PCBs and Wire Harnesses to acquire telemetry, engine, and car data to maximize performance while adhering to safety regulations

Rosendin | Engineer Intern/Co-Op | **May 2026 - Present**

- Conducted thorough short circuit and arc flash analysis for power distribution systems ensuring strict protocol and safety compliance with NEC and IEEE codes
- Planned system coordination and developed a disciplined approach to power management, fault isolation, and system reliability with ETAP and SKM

PROJECTS

Analog to Digital Converter Data Acquisition and Power Distribution Board, Mines Formula

- Architected a mixed-signal data acquisition board using an STM32 MCU and a dedicated 16-bit ADC communicating over SPI, implementing low-pass input filters for signal integrity
- Routed a power distribution network stepping a 12V input down through a buck converter and using LDO regulators to isolate 5V and 3.3V rails into analog and digital domains
- Engineered high-speed, impedance controlled differential pair traces for a 100Base-TX Ethernet PHY for reliable, low latency communication with the Vehicle Control Unit

Custom STM32 Microcontroller Development Board, Mines Formula

- Architected a 4-layer development board with pin exposure for Arduino Shield compatibility and testing via Serial Communications like CAN, SPI, I^2C, and UART
- Adhered to >80% power conversion efficiency (12V to 3.3V) using switching and linear regulators isolating the switching converter from sensitive analog traces to minimize EMI
- Routed differential pairs for USB 2.0 and CAN, while optimizing high-speed oscillator placement for minimizing EMI and maximizing clock stability

Power and Safety Circuit Latching Board, Mines Formula

- Designed an analog safety board utilizing relays, optocouplers, diodes, fuses, and MOSFETs to provide reliable voltage switching, galvanic isolation, and safe EV shutdown
- Engineered HV/LV safety architecture to reduce the Bill of Materials, validating fault detection logic via LTspice and testing with an oscilloscope and DC power supply